

How to access the DPU System

1. Connect to the cluster

When you receive confirmation, that your account has been activated, you can then connect to the cluster by opening a terminal and using the:

```
ssh user-name@172.25.1.11
```

2. Running Jobs on the DPU HPC Cluster

The queuing system used at DPU is Slurm. DPU system is a shared system, all computations must be carried out through a queue. Users submit jobs to the queue and the jobs then run when it's their turn. To handle different workloads, jobs can be

submitted to one or more partitions, which are essentially queues that have been assigned certain restrictions such as the maximum running time, maximum number of jobs, maximum number of GPUs, etc.

```
[myuser@login]$ sinfo
```

This will list each partition, all of the compute nodes assigned to each partition and the maximum wall time (running time) a job in the partition can have. By typing a slightly different command:

```
[myuser@headnode1]$ sinfo -N --long
```

All nodes will be listed and all partitions they belong to, together with the available resources such as the number of cores per node, and available memory per node. The queuing system allows us to submit a batch job.

3. Applications

Several programs with different versions are available on DPU systems. To use python3.10 the appropriate module should be loaded.

```
$module load python/3.10.gpu
```

Modules are “configuration” set/change/remove environment variables from the current environment. The useful module commands

module avail: display the available packages that can be loaded
module list: lists the loaded packages
module load foo: to load the package foo
module rm foo: to unload the package foo
module purge: to unload all the loaded packages

Module Description:

- a. python/3.10.gpu : having tensorflow and its supported libraries.
- b. python/3.10.pytorch : having pytorch and its supported libraries.

4. Batch jobs file example.

Batch jobs are best suited for computations lasting longer or resource-demanding analyses. Batch jobs are submitted to the queue like interactive jobs, but they don't give you a shell to run commands. Instead, you must write a job script that contains the commands that need to be run.

```
#!/bin/bash
# Job name:
#SBATCH --job-name=test

# Partition:
#SBATCH --partition=small<use appropriate partition name>
#SBATCH --nodes=1

#SBATCH --ntasks=1
## Processors per task:

#SBATCH --cpus-per-task=2
#
#SBATCH --gres=gpu:1 ##Define number of GPUs
```

```
#  
## Command(s) to run (example):  
module load python/3.10.gpu  
mpirun python3 /csehome/test/tr.py
```

→ To submit a job for this script, save it to a file (e.g. test.sh) and run:

```
[myuser@login]$ sbatch test.sh  
Submitted batch job 191  
[myuser@login]$
```

- Checking job status

→ To check the status of a job:

```
[myuser@login]$ squeue -j 191
```

→ To check the status of all of your submitted jobs:

```
[myuser@login]$ squeue -u USERNAME
```

→ You can also omit the username flag to get an overview of all jobs that have been submitted to the queue:

```
[myuser@login]$ squeue
```

→ **Cancelling a job**

Jobs can be canceled using the **scancel** command:

```
[myuser@login]$ scancel 191
```